Set Top Box Multiplayer Framework

Team: Plugged-In
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Sponsored by: Innovative Systems, L.L.C

Client

- Innovative Systems
  - Provides integrated solutions for the independent telecommunications industry.
- APMAX
  - Application Server providing services such as:
    - IPTV Video Solutions
    - Unified Messaging
    - Single Number Service
- STB
  - Set-top-box providing television and integrated telephony applications

Project Overview

- Modify the existing STB architecture to allow multiplayer games to be played
- Create a server to host game plugins and communication between clients
- Write a set of protocols that allow a generalized subset of games to be played
  - *i.e. board games, card games, etc…*

Project Overview (cont..)

- Implement a temporary pass-through messaging middle layer to simulate and extend the current in place system.
- Choose a board game and using our own framework, create a fully functional multiplayer version as a proof of concept

Last Semester

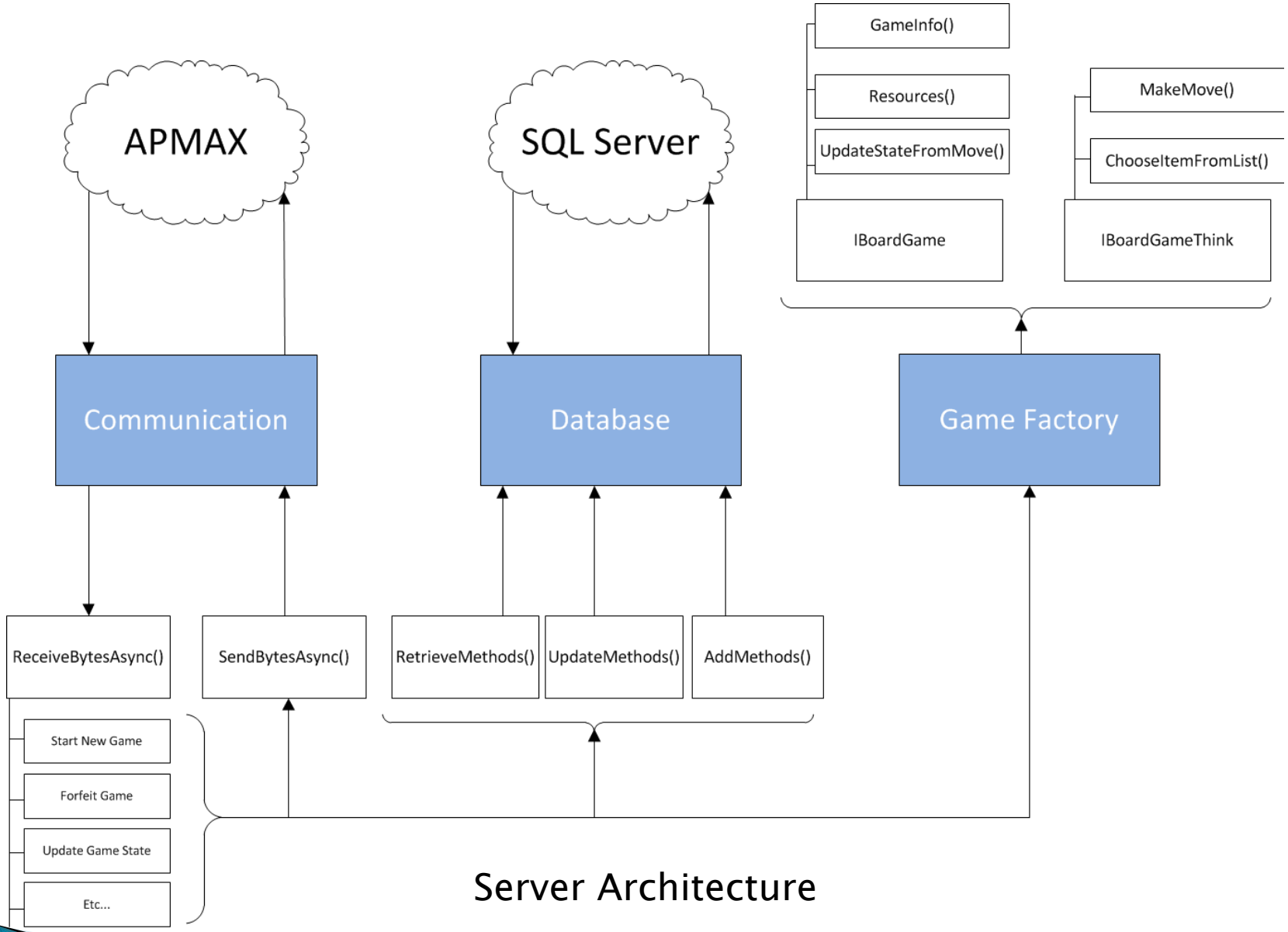
- Collected user stories
- Designed protocols
- Researched tools
- Implemented virtual APMAX

This Semester

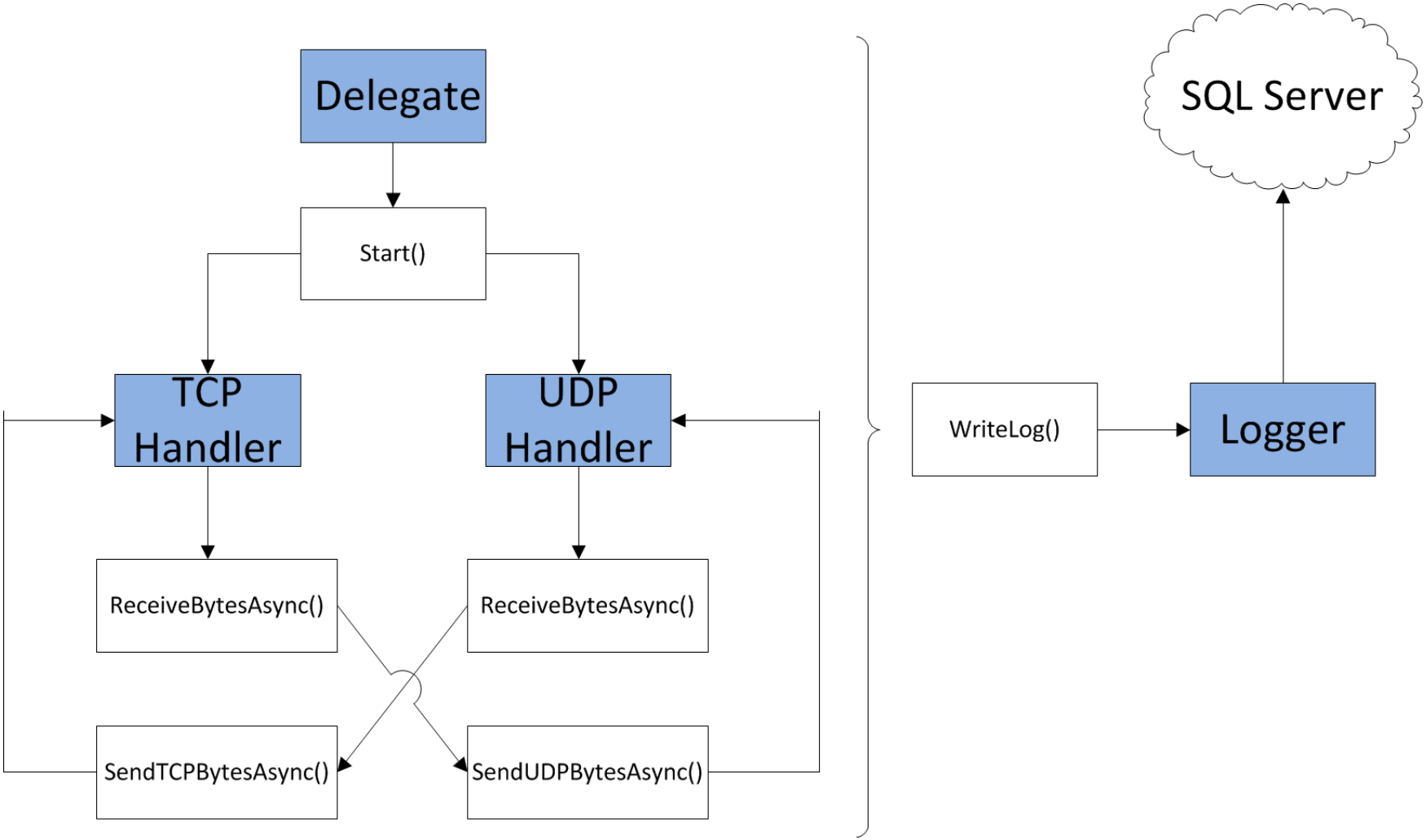
- Sprint 4
  - Designed server architecture
  - Designed vSTB layout
  - Created demo to show client
- Sprint 5
  - Implemented server and began unit testing
  - Created checkers logic and AI
  - Refactored vSTB

Remaining Work

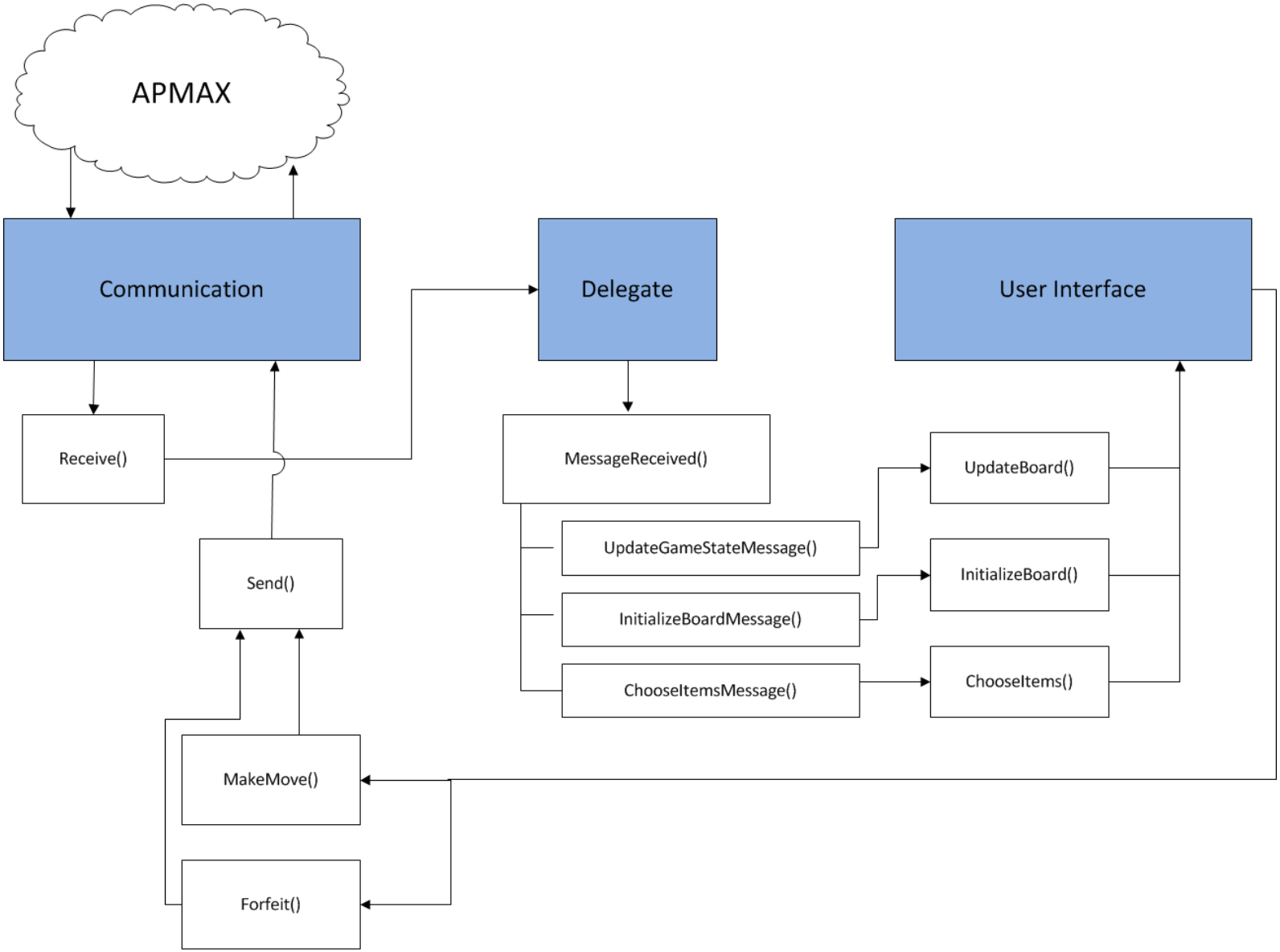
- Server
  - Finish unit testing
  - Create load testing environment
  - Documentation
    - Current system
    - Unimplemented features
- vSTB
  - Finish remote control functionality
  - Begin unit testing
  - Documentation
- Checkers
  - Optimize AI
  - Documentation



Server Architecture



vAPMAX Architecture



vSTB Architecture

Checkers

- Human vs. Human
  - Validate move
  - Return updated game
- Human vs. AI
  - Using MiniMax
  - 4 move look-ahead (2 ply)
- Improvements to AI
  - Alpha-Beta Pruning
    - Allow larger look-ahead
  - AI specific move validation
    - Existing validation proving too bulky

Testing Plan

- Server
  - Unit Testing
    - Moq library
    - Dependency injection
  - Load Testing
    - Response time
    - Reduce bottlenecks
- vSTB
  - QTest vs QTestLib
- vAPMAX
- Checkers

Demo

Questions?